DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) — ROUND 5

SentiLingo

Multilingual Customer Sentiment Analysis Platform



Project Documentation

Part 1 Project Planning & Management

Part 2 Literature Review

Part 3 Requirements Gathering

Track	Microsoft Machine Learning
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Version	1.1 — 2 November 2026

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Document control

Document	SentiLingo Project Documentation — Parts 1–3 · version 1.1 · 2 November 2026 (deadline 2 November 2026)
Prepared by	SentiLingo project team
Next revision	Remaining documentation parts, following the DEPI *Project Documentation Guidelines* schedule

PART 1

1. Project Planning & Management

What SentiLingo is, what it will deliver and when, how the team works, which risks matter most and how success is measured.

1.1 Project Overview

Companies receive customer feedback through app and e-commerce reviews, social media, surveys and support tickets. In Egypt this feedback is commonly written in Modern Standard Arabic (MSA), Egyptian Arabic and English, and may mix Arabic and English in one message. Tagging it by hand is slow and inconsistent, and many ready-made tools do not report how well they handle Egyptian Arabic (see Part 2).

> **Problem statement (from the project brief).** A company receives customer feedback in natural-language text and needs to automatically identify whether feedback expresses positive, negative, or other predefined sentiment categories.

SentiLingo labels each piece of feedback **Negative**, **Neutral** or **Positive** with an estimated confidence. It compares a dummy classifier and focused classical baselines using word, character and combined TF-IDF with Multinomial Naive Bayes, Logistic Regression and Linear SVM. XLM-R Base is the required advanced model; MARBERT is an Arabic-only stretch experiment and BiLSTM is included only if the instructor requires it. Models are selected using the mean of the English, MSA and Egyptian Arabic macro-F1 scores on validation data. The locked configuration is then evaluated once on the frozen test set.

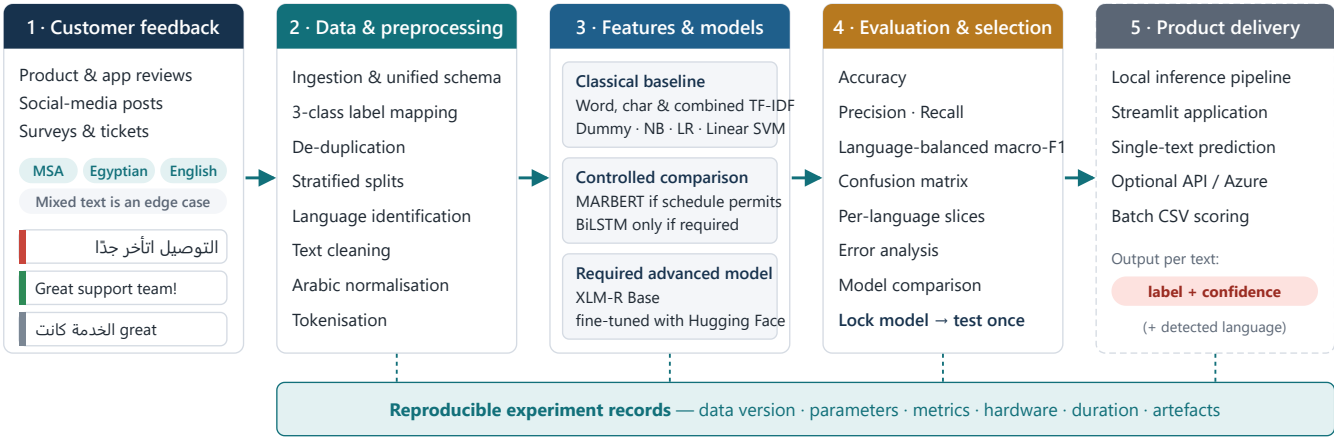


Figure 1 — SentiLingo solution overview.

1.2 Objectives

ID	Objective	Due (2026)
O1	Approve project planning, stakeholder, database and UI/UX design artefacts; select licensed primary and backup datasets.	8 Nov
O2	Build a unified English, MSA and Egyptian Arabic dataset with three sentiment classes, frozen splits and a data card.	15 Nov
O3	Compare word, character and combined TF-IDF using a dummy classifier, Multinomial Naive Bayes, Logistic Regression and Linear SVM.	15 Nov
O4	Fine-tune XLM-R Base; attempt MARBERT or BiLSTM only under the documented stop rule.	22 Nov
O5	Select the model on validation data, evaluate the locked configuration once on the test set and complete error analysis.	29 Nov
O6	Deliver a reliable local Streamlit application for single-text and CSV prediction.	29 Nov
O7	Deliver the repository, documentation, model card, presentation PDF, video demonstration and offline backup.	4 Dec

1.3 Scope

In scope

- English, Modern Standard Arabic and Egyptian Arabic
- Three classes: Negative, Neutral, Positive
- Public, labelled review and tweet data
- Word, character and combined TF-IDF with dummy, NB, LR and Linear SVM baselines
- XLM-R Base, reproducible experiment records, evaluation and error analysis
- Local Streamlit application with single-text and CSV prediction

Out of scope

- Aspect-based sentiment and emotion detection
- Speech, image or video feedback
- Live social-media crawling and CRM integration
- Collecting private customer data
- Production SLAs and automated retraining
- Franco-Arabic and Arabic dialects other than Egyptian Arabic
- User accounts, authentication and a production database
- FastAPI, Docker and Azure unless the core release is complete

Assumptions and constraints

- The five official team members have limited availability, so the plan protects a small required release and uses one owner and one reviewer for each deliverable.
- Zero budget: local resources, free Colab or Kaggle GPUs, and open-source tools; Azure remains optional.
- Only public data is used, and no personal data is committed to the repository.
- The local RTX 4050 with 6 GB VRAM is used for smoke tests and inference; hosted GPUs are used for the main Transformer runs.
- The active five-week delivery period starts on 2 November and the final presentation is on 4 December 2026.

1.4 Deliverables

ID	Deliverable (from the brief)	Description	Milestone
D1	Planning and design package	Project planning, stakeholder analysis, database design and UI/UX design	MS0 · 2 Nov
D2	Dataset and data card	Source register, licences, label mapping, common schema, quality audit and frozen splits	MS2 · 15 Nov
D3	NLP preprocessing	Reproducible cleaning, Arabic normalisation and feature preparation with leakage checks	MS2 · 15 Nov
D4	Classical baselines	Dummy, NB, LR and Linear SVM with word, character and combined TF-IDF	MS2 · 15 Nov
D5	Advanced model	XLM-R Base plus an optional controlled MARBERT or BiLSTM comparison	MS3 · 22 Nov
D6	Evaluation and error analysis	Validation leaderboard, one locked test evaluation, confusion matrices and error taxonomy	MS4 · 29 Nov
D7	Local application	Streamlit single-text and CSV prediction with Arabic right-to-left display	MS4 · 29 Nov
D8	Reproducibility package	Experiment records, tests, environment instructions, data card and model card	MS5 · 4 Dec
D9	Final documentation	README and consolidated technical documentation with decisions and limitations	MS5 · 4 Dec
D10	Presentation package	Presentation PDF, video demonstration, rehearsal and offline backup	MS5 · 4 Dec

1.5 Methodology

The project follows CRISP-DM (Wirth & Hipp, 2000): business understanding, data understanding, data preparation, modelling, evaluation and deployment. The work is compressed into five weekly delivery cycles. Each cycle begins with a short planning meeting, assigns one owner and one reviewer to every deliverable, and ends with evidence-

based review. GitHub issues track work, reviewed pull requests protect the main branch, and every reported result records its data version, configuration, seed, hardware, duration, metrics and artefact location.

1.6 Timeline & Milestones

The active delivery plan runs from **2 November** to the final presentation on **4 December 2026**, in five focused weeks (W1–W5). The final date is fixed; internal milestones are stop points that prevent optional work from displacing the required release.

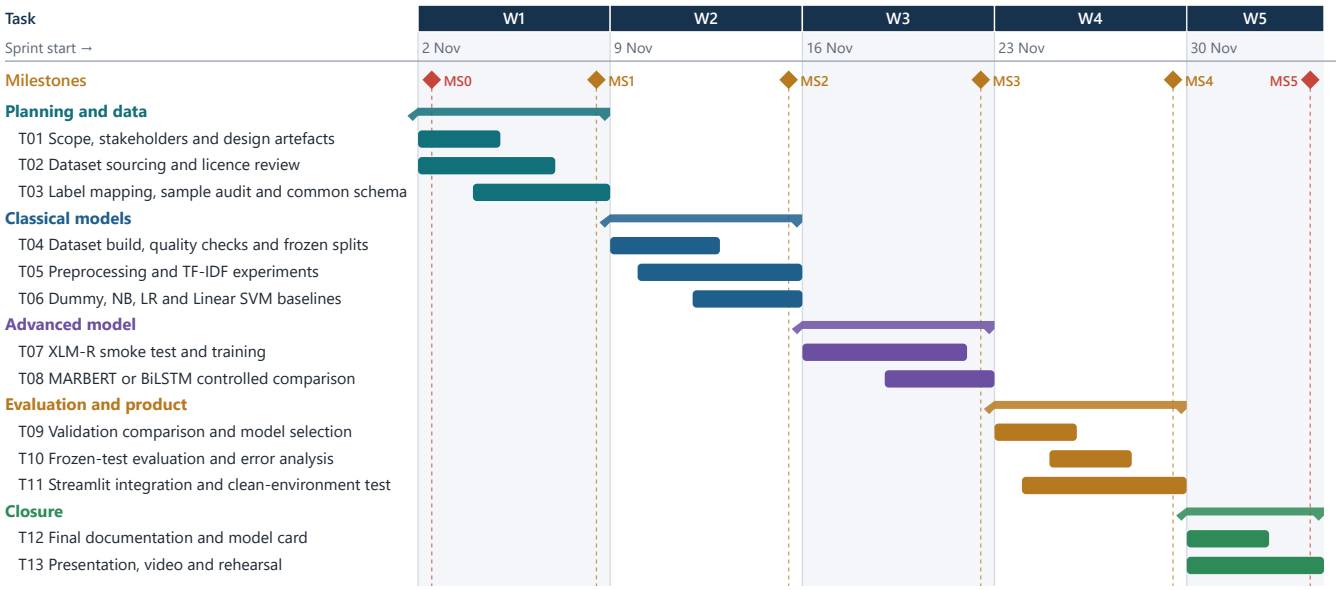


Figure 2 — Project schedule. Bars are coloured by phase; ◆ marks a milestone (red = fixed date).

ID	Milestone	Target date	Exit criteria (evidence)	Type
MS0	**Planning package and repository aligned**	2 Nov 2026	Project plan, stakeholders, database design, UI/UX design and repository links are prepared	Official DEPI
MS1	**Data decisions approved**	8 Nov 2026	Primary and backup sources, licences, label mapping, common schema and sample audit are approved	Internal
MS2	**Dataset and classical baselines ready**	15 Nov 2026	Frozen splits, data card and comparable dummy, NB, LR and Linear SVM results exist	Internal
MS3	**Advanced model ready**	22 Nov 2026	XLM-R is trained and evaluated on validation data; optional comparison is either completed or formally dropped	Internal
MS4	**Evaluation and local demo ready**	29 Nov 2026	Model is locked, test evaluation is complete, errors are analysed and the Streamlit demo works locally	Internal
MS5	**Final presentation**	4 Dec 2026	Documentation, repository, PDF presentation, video demonstration and offline backup are delivered	Official DEPI

1.7 Team & Task Distribution

The team has five members:

1. احمد مؤنس احمد عبدالفتاح
2. نوران ناصر جلال سيد
3. محمد محمود محمد عبدالله عياد
4. احمد اشرف سامى فريد
5. رضوه أشرف جمعه مهدي

The official team has five members. Work is split into five accountable areas, but ownership may rotate as availability changes. Every deliverable has one owner, one reviewer and a visible completion criterion. No task is considered complete only because a file exists.

Work area	Main tasks
Product and planning	Scope, stakeholders, backlog, risks, design decisions and delivery evidence
Data and quality	Dataset sourcing, licence checks, label mapping, sample audit, schema, splits and data card
Classical modelling	Preprocessing, word and character TF-IDF, dummy, NB, LR and Linear SVM experiments
Advanced modelling and evaluation	XLM-R training, optional comparison, experiment records, model selection and error analysis
Application and communication	Streamlit, tests, README, technical report, presentation, video and rehearsal

Communication: one weekly planning and review meeting, brief asynchronous status updates, GitHub issues for ownership and acceptance criteria, and pull requests for review. Optional tasks require explicit approval after required work is on schedule.

1.8 Risk Management

Risks are reviewed at every sprint review. The level combines likelihood and impact before mitigation.

ID	Risk	Level	Mitigation
R1	Too little labelled Egyptian-dialect customer feedback	High	Combine several public Arabic sources; label a small gold set by hand; report the dialect slice separately
R2	A dataset is withdrawn or has a restrictive licence	High	Check licences in week 1; keep backup sources; commit download scripts, not raw data
R3	Limited GPU memory, hosted-GPU quota or interrupted sessions	High	Use local CPU for classical work, local GPU for smoke tests and inference, hosted GPU for main runs, small batches and saved checkpoints
R4	Data leakage: duplicates across splits, or preprocessing fitted on all data	High	De-duplicate before splitting; scikit-learn Pipelines fitted on training data only; frozen test set
R5	Inconsistent labels across sources, and too few Neutral examples	Medium	Written label-mapping policy; manual audit; stratified splits; class weights; macro-F1 as the main metric
R6	Egyptian spelling variation, mixed text, sarcasm and negation cause errors	Medium	Arabic normalisation, character n-grams, language-specific evaluation, difficult-example audit and explicit limitations
R7	Uneven availability of team members and delayed reviews	High	One owner and one reviewer per deliverable, small tasks, early escalation and required-work stop rules
R8	Optional API, Docker, Azure or extra models consume the final week	Medium	Keep Streamlit local-first and start optional work only after evaluation, documentation and rehearsal are safe

1.9 Key Performance Indicators

ID	Key performance indicator	Success rule
K1	Primary validation score	**Mean of English, MSA and Egyptian Arabic macro-F1**; no language may be hidden by the largest subset
K2	Evidence over a trivial baseline	The selected model must beat the dummy classifier and be compared fairly with the strongest classical baseline
K3	Per-language and per-class transparency	Macro-F1, class metrics and confusion matrices are reported for all three supported varieties
K4	Prediction usability	Single-text and CSV prediction complete reliably on the presentation laptop; latency and model size are recorded
K5	Experiment traceability	**100%** of reported results identify data, configuration, seed, hardware, duration and artefacts
K6	Required milestone protection	Required work is completed before any optional model or cloud deployment begins
K7	Delivery completeness	Repository, documentation, presentation PDF, video and offline demo backup all open from the submitted links

Validation data selects the model; the frozen test set is evaluated once after all preprocessing, model and threshold decisions are locked.

PART 2

2. Literature Review

Related research and existing solutions, what SentiLingo borrows from them, and where it differs.

2.1 Review Method

We searched the ACL Anthology, Google Scholar, arXiv and vendor documentation for work on the sentiment of reviews and social-media text, focusing on Arabic, Egyptian dialect and multilingual models. Every figure quoted below was checked against the original source (October 2026).

2.2 Background and Challenges

Sentiment analysis classifies the opinion expressed in a text. Customer feedback makes this hard: messages are short and informal, often mix praise and complaints, and neutral messages are hard to label consistently. Arabic adds further difficulties:

- rich morphology and inconsistent spelling (e.g. ا/أ, ي/ى);
- dialects such as Egyptian Arabic that differ from MSA in vocabulary and grammar;
- Franco-Arabic and code-switching with English, for which very few labelled resources exist (Hamed et al., 2025);
- sarcasm and label noise, the main sources of the errors that remain in Arabic sentiment models (Abu Farha & Magdy, 2021).

2.3 Classical Approaches

Lexicon-based tools such as VADER score text with a hand-built word list plus rules for negation and intensity. They need no training data but are English-only (Hutto & Gilbert, 2014).

Supervised learning does better:

- On movie reviews, Naive Bayes and SVM classifiers on word features reached 81–83% accuracy and beat keyword lists (Pang et al., 2002).
- Adding word bigrams lifted a Naive Bayes–SVM hybrid to 91.2% (Wang & Manning, 2012).
- For Arabic, an SVM on TF-IDF n-grams scored 0.63 F1 on the four-class Egyptian tweet corpus ASTD (Nabil et al., 2015).
- The best Arabic system at SemEval-2017 was a Naive Bayes model (Rosenthal et al., 2017).

These models are fast, cheap, interpretable and effective with modest data. However, they see no context beyond n-grams, are weak on negation and sarcasm, and break on unseen dialect spellings. This is the baseline required by the brief; SentiLingo adds character n-grams to handle spelling variation.

2.4 Deep Learning and Transformers

LSTMs (Hochreiter & Schmidhuber, 1997) read text as a sequence and can learn effects such as the scope of a negation. They need more data and tuning than linear models.

Pretrained Transformers are first trained on large text corpora and then fine-tuned for the task:

- BERT reached 94.9% on a standard English sentiment benchmark (Devlin et al., 2019).
- XLM-R extends this approach to 100 languages (Conneau et al., 2020).
- XLM-T, a Twitter-adapted version of XLM-R, reached macro-F1 66.9 on Arabic and 70.6 on English three-class tweets (Barbieri et al., 2022).

Arabic-specific models do better on Arabic:

- AraBERT reached 92.6% accuracy on the balanced ASTD, against 80.1% for multilingual BERT (Antoun et al., 2020).

- MARBERT, pretrained on one billion dialect-rich tweets, beat both XLM-R and multilingual BERT on SemEval-2017 Arabic (Abdul-Mageed et al., 2021).

Large language models can classify sentiment without any training. They do well on simple tasks but trail fine-tuned smaller models on harder ones, and they cost much more per prediction (Zhang et al., 2024).

2.5 Arabic and Egyptian-Dialect Studies

The study closest to SentiLingo compared a TF-IDF SVM, an LSTM, multilingual BERT and Arabic BERT models on 40,000 Egyptian tweets (Abu Kwaik et al., 2022). The SVM (0.84 accuracy) matched multilingual BERT (0.83), a Twitter-adapted Arabic BERT was best (0.91) and the LSTM was weakest (0.49); the study used two classes only. A re-implementation of published Arabic methods found that Transformers gave the best F-scores: 0.69 on SemEval-2017 and 0.76 on ASTD (Abu Farha & Magdy, 2021). For code-switched feedback, a corpus of 4,100 Egyptian Arabic–English comments reached an F1 of about 0.93, both with neural models and with a fine-tuned GPT-3.5 (Sherif & Sabty, 2025).

2.6 Existing Solutions

Solution	Type	Arabic support	Strengths	Weaknesses
Azure Language sentiment analysis (Microsoft, 2026)	Cloud API	Generic Arabic	Managed; labels with confidence per sentence and per document	Cannot be customised to the user's data; the feature retires on 31 Mar 2029
Amazon Comprehend (Amazon Web Services, n.d.)	Cloud API	Generic Arabic	Managed; batch jobs; four labels including Mixed	Entity-level sentiment in English only; pay per use
Google Cloud Natural Language (Google Cloud, n.d.)	Cloud API	Generic Arabic	Simple score and magnitude for each sentence	No class labels, so thresholds must be set; no customisation within the API
VADER (Hutto & Gilbert, 2014)	Open-source lexicon	None	Free, fast, no training needed	English only; misses domain and dialect words
Twitter-XLM-R sentiment model (Barbieri et al., 2022)	Open pretrained model	8 languages, including Arabic	One multilingual model, ready to use	Trained on tweets; about 0.67 macro-F1 on Arabic

Table 2.1 — Ready-made options. None of the cloud APIs lists Egyptian Arabic or Franco-Arabic as a supported variety.

2.7 Gap Analysis

#	Gap in existing work	SentiLingo response
G1	Egyptian dialect and Franco-Arabic are poorly covered by ready-made tools	One pipeline for MSA, Egyptian Arabic, English and Franco-Arabic, evaluated per language
G2	Classical and Transformer models are rarely compared on the same data; studies often use one dataset and two classes	The same frozen splits, seeds and metrics for every model family, with three classes
G3	Commercial APIs cannot be customised and can be retired	An owned model that can be retrained on company data
G4	Published results are hard to reproduce	Every run tracked in MLflow, with reproduction steps in the README
G5	LLMs are costly per prediction	Fine-tuned compact models, with latency reported next to accuracy

2.8 Lessons Applied

1. Tune the classical baseline carefully, with word and character n-grams, and keep negation words.
2. Compare a multilingual Transformer with a dialect-aware Arabic model.
3. Report macro-F1 and per-class metrics, with realistic targets: about 0.67–0.71 macro-F1 is already strong for three-class tweets.

PART 3

3. Requirements Gathering

Who SentiLingo serves, what it must do, how well it must do it, and the data and tools it needs.

3.1 Elicitation Approach

Requirements were gathered from five sources:

1. the DEPI project brief, which defines the mandatory scope;
2. the review of existing solutions in Part 2;
3. a team workshop in which members played the stakeholders listed below;
4. inspection of the candidate datasets;
5. guidance from the technical instructor.

Each requirement has an ID and a MoSCoW priority: Must, Should or Could.

3.2 Stakeholders & Their Needs

Stakeholder	Key needs
Customer-experience (CX) managers	Reliable sentiment figures per language and channel; batch analysis; results they can export
Support teams	Negative feedback surfaced first, with a confidence score; works for Egyptian Arabic and Franco-Arabic
Data scientists / ML engineers	A reproducible, configurable pipeline; tracked experiments; easy retraining
IT and security teams	Secure, low-cost deployment; no leakage of customer data
Management	Evidence that the chosen model is accurate and cost-effective
DEPI instructor and evaluators	All brief deliverables, a sound method, and documentation delivered on time

3.3 User Stories

ID	User story	Acceptance criteria
US-01	As a **CX manager**, I want to upload a CSV of feedback and get a sentiment label and confidence for every row.	10,000 rows processed; the output adds label, confidence and language.
US-02	As a **support agent**, I want to paste one message in Arabic, English or Franco-Arabic and see its sentiment.	Result in under 1 second; Arabic shown right-to-left.
US-03	As a **data scientist**, I want every run logged with its parameters, metrics and artefacts.	Each MLflow run holds all brief metrics and a confusion matrix.
US-04	As a **manager**, I want all models compared on accuracy and speed.	A leaderboard shows macro-F1, per-language F1 and latency.
US-05	As an **evaluator**, I want to reproduce the reported results.	Following the README from a fresh clone reproduces the test metrics.

3.4 Functional Requirements

ID	Requirement	Priority
FR-01	Load labelled data from several sources into one schema: text, label, language, source, split.	Must
FR-02	Map each source's labels or star ratings to Negative, Neutral and Positive using a documented policy.	Must
FR-03	Remove duplicates, then create stratified train / validation / test splits with a fixed seed. The test set is frozen.	Must
FR-04	Detect the language or script of each text: Arabic, English or Franco-Arabic.	Should
FR-05	Clean text: mask URLs, mentions, e-mails and phone numbers; strip HTML; handle emojis; lower-case Latin script.	Must

ID	Requirement	Priority
FR-06	Normalise Arabic: remove diacritics and tatweel; unify alef forms (آ إ أ → ا) and ى → ي; reduce repeated letters.	Must
FR-07	Tokenise text: word level for the classical and LSTM models, subwords for Transformers. Negation words are kept.	Must
FR-08	Build TF-IDF features with word and character n-grams.	Must
FR-09	Train Naive Bayes and Logistic Regression with cross-validated tuning.	Must
FR-10	Train a BiLSTM classifier.	Must
FR-11	Fine-tune at least one pretrained Transformer, multilingual and/or Arabic.	Must
FR-12	Report accuracy, precision, recall, F1 (macro and per class) and the confusion matrix, overall and per language.	Must
FR-13	Compare preprocessing strategies and models in one leaderboard.	Must
FR-14	Log every run to MLflow and register the best model.	Must
FR-15	Produce an error analysis and a table of sample predictions.	Must
FR-16	Predict single texts and CSV files, returning a label and a confidence score.	Should
FR-17	Provide a Streamlit demo that shows Arabic right-to-left.	Should
FR-18	Serve the best model from an Azure ML online endpoint (REST, key authentication).	Could
FR-19	Provide a README and a model card.	Must

3.5 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The selected model meets KPIs K1–K3 (target macro-F1 ≥ 0.75).
NFR-02	Performance	One text is scored on CPU in ≤ 20 ms (baseline) or ≤ 300 ms (Transformer).
NFR-03	Security	No API keys or tokens in the repository; secrets come from environment variables.
NFR-04	Privacy	Personal data (handles, e-mails, phone numbers) is masked, and raw third-party data is never committed.
NFR-05	Usability	A first-time user can score a CSV file in under 2 minutes; Arabic is displayed right-to-left.
NFR-06	Reproducibility	Fixed seeds, pinned dependencies and versioned splits; a re-run reproduces macro-F1 within ±0.01.
NFR-07	Reliability	Empty, emoji-only, mixed-script or very long inputs never crash the pipeline.
NFR-08	Maintainability	Modular code in `src/`, with unit tests for preprocessing and evaluation.
NFR-09	Fairness	Results are reported per language, and the Arabic–English macro-F1 gap stays within 10 points.
NFR-10	Compliance	Datasets and models are used under their licences, and all sources are cited.

3.6 Data Requirements

Label	Definition	English example	Egyptian Arabic example	Franco-Arabic example
Negative	Dissatisfaction, a complaint or criticism	"The order arrived broken."	الطلب وصل مكسور	"el order wesel maksour"
Neutral	No clear sentiment: facts, questions or balanced opinions	"When does the store open?"	المحل بيفتح الساعة كام؟	"el ma7al beyefta7 emta?"
Positive	Satisfaction, praise or a recommendation	"Fast delivery, thank you!"	التوصيل سريع، شكرًا	"el delivery saree3, shokran"

Label mapping: star ratings 1–2 → Negative, 3 → Neutral, 4–5 → Positive. Source labels such as *objective* or *mixed* map to Neutral, and mixed items are flagged so they can be analysed separately.

Candidate dataset	Language	Domain	Size	Licence	Planned use
HARD (Elnagar et al., 2018)	MSA and dialects	Hotel reviews	≈ 374k reviews, rated 1–5	None declared	Main Arabic training data
ASTD (Nabil et al., 2015)	Egyptian and MSA	Tweets	10,006	GPL-2.0	Egyptian-dialect training data
Twitter US Airline Sentiment (CrowdFlower, n.d.)	English	Customer-service tweets	14,640	CC BY-NC-SA 4.0	Main English training data
SemEval-2017 Task 4A (Rosenthal et al., 2017)	Arabic, English	Tweets	9,455 Arabic; 62,617 English	Free use beyond SemEval	Extra three-class training data and a benchmark
UMSAB (Barbieri et al., 2022)	8 languages, including Arabic and English	Tweets	3,030 per language	CC BY 3.0	External multilingual benchmark
JERD — Jumia Egypt Reviews (aliallam13, 2026)	Egyptian, MSA, English, Franco-Arabic	E-commerce reviews	3,809	CC BY-NC-ND 4.0	Held-out test set of real Egyptian feedback

- At least 10,000 labelled texts per core language after cleaning, and every test slice contains all three classes.
- Splits are 70 / 15 / 15, stratified by language and label. The test set is frozen at MS1.
- Duplicates are removed before splitting, licences are recorded for each source, and raw data is not committed.

3.7 Technology Stack

Layer	Tools
Language and environment	Python 3.10+, Jupyter, Google Colab, Kaggle
Data and preprocessing	pandas, Hugging Face Datasets, regular expressions, NLTK, Arabic normalisation utilities
Classical ML	scikit-learn: TF-IDF, Naive Bayes, Logistic Regression, Pipelines
Deep learning	TensorFlow / Keras for the BiLSTM; Hugging Face Transformers on PyTorch
Experiment tracking	MLflow (Zaharia et al., 2018)
Serving and collaboration	Azure Machine Learning, Streamlit; GitHub Issues, Projects and pull requests

3.8 Use-Case Model

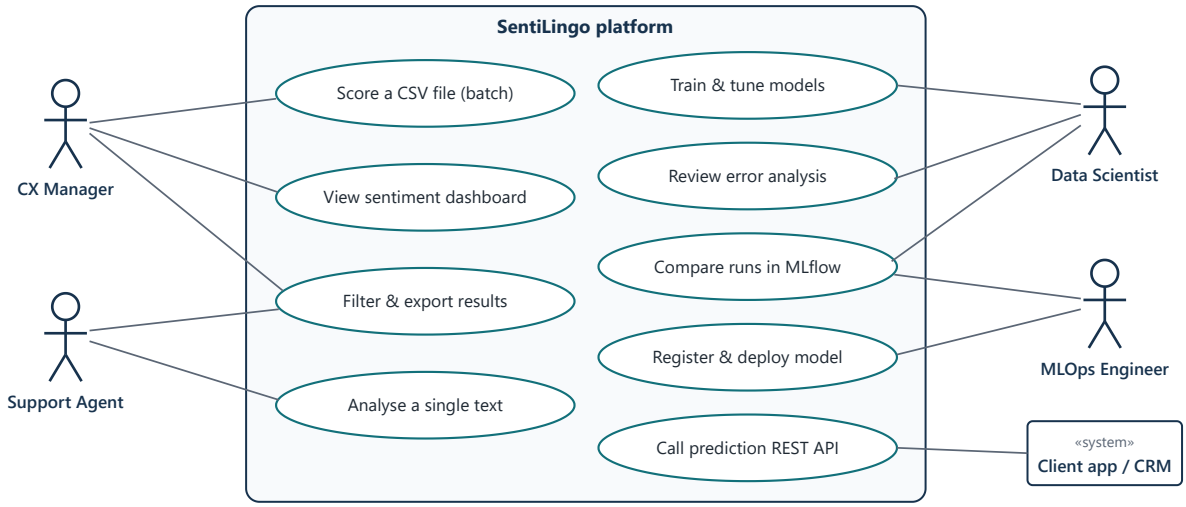


Figure 3 — Use-case diagram: business users (left) consume predictions; technical users and external systems (right) build, deploy and integrate the model.

3.9 Traceability to the Brief

Brief item	Requirements	Deliverable
Dataset	FR-01 – FR-04	D1
Text cleaning, tokenisation	FR-05 – FR-07	D2
TF-IDF, n-grams, Naive Bayes / Logistic Regression	FR-08, FR-09	D3
RNN / LSTM or pretrained Hugging Face Transformer	FR-10, FR-11	D4
Accuracy, precision, recall, F1, confusion matrix	FR-12	D5
Compare preprocessing strategies and models	FR-13	D5
Track experiments using MLflow	FR-14	D6
Error analysis, sample predictions	FR-15	D7, D8
README	FR-19	D9
Optional: Azure ML endpoint	FR-18	D10

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APPENDIX

Appendix A — Glossary

Term	Meaning in this document
Franco-Arabic (Arabizi)	Arabic written in Latin letters, with digits standing for Arabic sounds (e.g. 3 = ع, 7 = ح), as in `el service 7elw`.
Code-switching	Mixing two or more languages or scripts in one message.
MSA / Egyptian Arabic	Modern Standard Arabic, the formal written variety / the everyday Arabic dialect of Egypt.
TF-IDF	Term frequency–inverse document frequency: weights a word by how often it appears in a text, discounted by how common it is across all texts.
N-gram	A sequence of *n* consecutive words or characters, used as a feature.
BiLSTM	A bidirectional Long Short-Term Memory network: a recurrent model that reads text in both directions.
Transformer / fine-tuning	A neural architecture based on self-attention / further training a pretrained model on labelled task data.
Macro-F1	The average of the per-class F1 scores; every class counts equally, so minority classes are not ignored.
Confusion matrix	A table of predicted against true classes, showing which classes the model confuses.
MLflow	An open-source platform for tracking experiments and registering models.
CRISP-DM	A standard data-mining process: business understanding, data understanding, data preparation, modelling, evaluation, deployment.
MoSCoW	A prioritisation scheme: Must, Should, Could, Won't (this time).